

User Manual

PathHunter® Checkpoint Assay - TNF Receptor Superfamily (TNFRSF)

For Chemiluminescent Detection of TNFRSF Checkpoint
Receptor Signaling

Table of Contents

Overview	1
Technology Principle	1
Materials Provided	2
Storage Conditions.....	2
Additional Materials Required	2
Cell Vials Handling Procedure	3
Cell Culture Protocol	3
Cell Thawing	3
Cell Propagation	4
Cell Freezing.....	5
TNFRSF Checkpoint Signaling Assay Protocol Schematic	7
Detailed Assay Protocol for Soluble Ligand or Antibodies.....	8
Step 1: TNFRSF Signaling Cell Preparation and Plating.....	8
Step 2: Soluble Ligand or Antibody Preparation	9
Step 3: Ligand Addition	10
Step 4: Detection Reagent Addition	10
Step 5: Assay Plate Reading.....	10
Representative Assay Plate Map	10
Typical Results for Soluble Ligand.....	11
Detailed Assay Protocol for Cross-linking Agonist Antibodies.....	12
Step 1: TNFRSF Signaling Cell Preparation and Plating.....	12
Step 2: Agonist Antibody Preparation.....	12
Step 3: Agonist Antibody Addition.....	12
Step 4: Cross-linking Antibody Preparation	12
Step 5: Cross-linking Antibody Addition	12
Detailed Assay Protocol for Cross-linking Agonist Antibodies by Co-culture with FcγR-expressing Cells.....	13
Step 1: TNFRSF Signaling Cell Preparation and Plating	13
Step 2: Agonist Antibody Preparation.....	13
Step 3: Agonist Antibody Addition	13



Please read this entire user manual before proceeding with the assay.
For additional information or Technical Support, see contact information below.

Step 4: FcγR-expressing Cells Preparation	13
Step 5: FcγR-expressing Cells Addition	13
Typical Results for Cross-Linking Agonist Antibodies	14
Troubleshooting Guide.....	15
Limited Use License Agreement.....	15



Please read this entire user manual before proceeding with the assay.
For additional information or Technical Support, see contact information below.

Overview

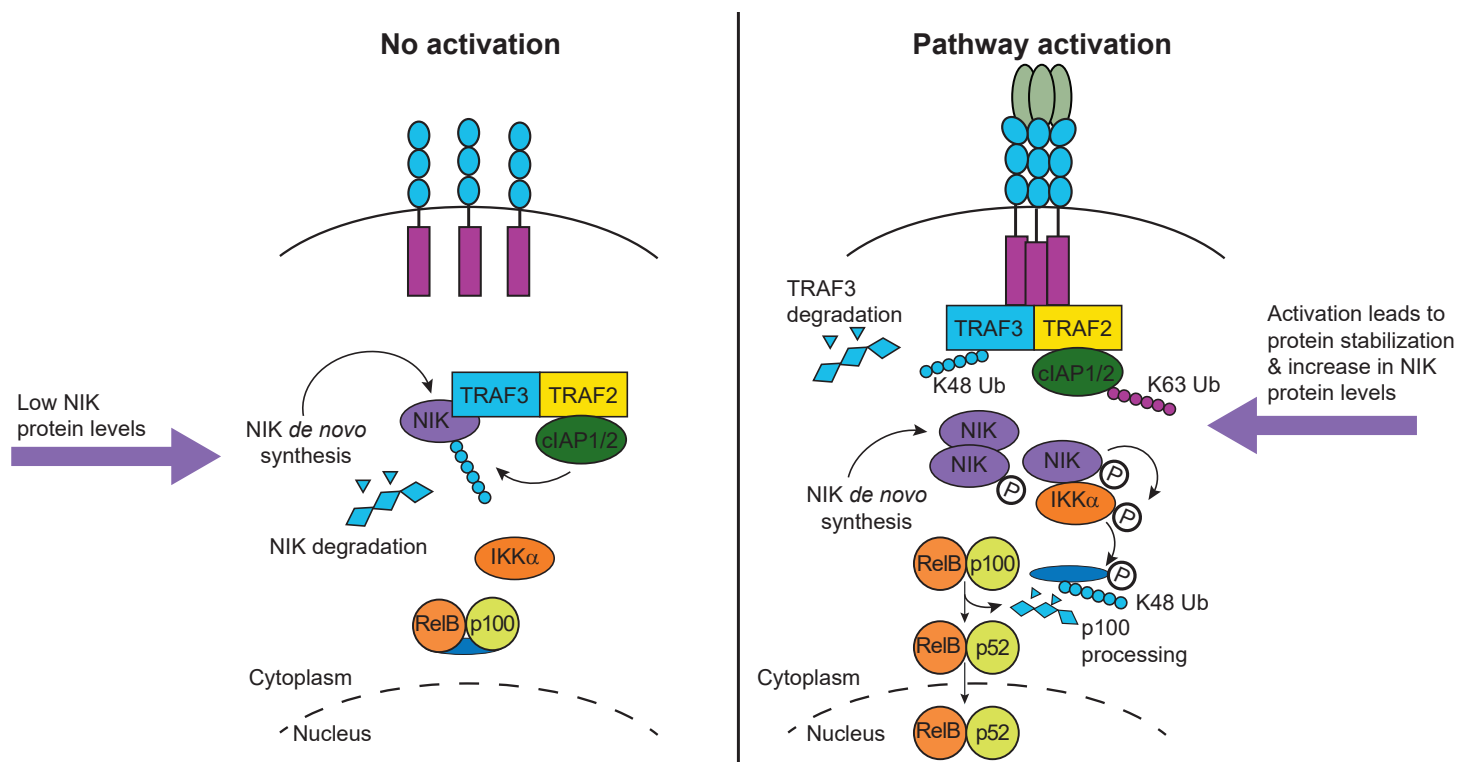
The PathHunter Checkpoint Signaling Assay for Tumor Necrosis Factor Receptor Superfamily (TNFRSF) members (or receptors) is a simple, highly specific assay to enable discovery and development of small molecule and biologic drugs targeting the TNFR superfamily of receptors that activate the non-canonical NF- κ B pathway through NF- κ B inducing kinase (NIK). The assay is a robust measurement of co-stimulatory checkpoint receptors using cells that express NIK. This has been developed for use with cell lines natively or exogenously expressing co-stimulatory receptors (e.g. cell lines engineered to express OX40 or CD137).

Technology Principle

The PathHunter TNFRSF Checkpoint Assay relies on the well-established PathHunter Enzyme Fragment Complementation (EFC) technology to interrogate receptor activity. EFC employs a split β -galactosidase (β -gal) enzyme, which is comprised of Enzyme Donor (ED) and Enzyme Acceptor (EA) fragments. Independently, these fragments have no β -gal activity. However, when forced to complement, they form an active β -gal enzyme that will hydrolyze substrate to produce a chemiluminescent signal.

Each TNFRSF Checkpoint Assay is a cell line engineered to express NIK-labeled with enzyme donor (ED) and an untagged, relevant receptor (i.e. OX40). NIK protein levels inside the cell are tightly maintained at low levels by the proteasome and increase in response to pathway activation. When cells are stimulated with ligand (i.e. OX40L), NIK is stabilized. NIK stabilization results in an increase in signal when detection reagent containing exogenous EA is added due to the forced complementation of the ED and EA enzyme fragments, resulting in the formation of a functional β -gal enzyme that hydrolyzes substrate to generate a chemiluminescent signal. This assay may rely on either endogenous receptors or exogenously expressed target receptors.

Principle of Checkpoint Signaling for TNF Receptor Superfamily (TNFRSF)



Materials Provided

Components	
2 vials of cells	Refer to cell line-specific datasheet for shipped cell density

Storage Conditions

Cells must arrive in a frozen state on dry ice, and should be transferred to the **vapor phase** of liquid nitrogen storage or a -80°C freezer **immediately upon arrival**. If cells will be thawed, and used within 24 hours, they can be stored temporarily at -80°C . For long-term storage, place vials in the vapor phase of liquid nitrogen storage

Additional Materials Required

Refer to the cell line-specific datasheet to determine catalog numbers for the media and reagent requirements for the specific cell line that you are testing.



It is highly recommended to not substitute CP Reagent.

Material	Ordering Information
Control Antibody	Refer to cell line-specific datasheet
AssayComplete™ Cell Culture Kit	Refer to cell line-specific datasheet
AssayComplete Cell Plating Reagent	Refer to cell line-specific datasheet
AssayComplete Cell Detachment Reagent	92-0009
AssayComplete Protein Dilution Buffer	92-0023 Series*
AssayComplete Thawing Reagent	Refer to cell line-specific datasheet
AssayComplete Freezing Reagent	Refer to cell line-specific datasheet
Detection Kit	Refer to cell line-specific datasheet
Ligand Cell Line (for co-culture assays only)	Refer to cell line-specific datasheet
96-well White Bottom TC Treated Plates, Sterile w/Lid, 10/ pack	92-0027
V-bottom PP Dilution Plates, 10/pack (for antibody dilutions)	92-0011
Disposable Reagent Reservoir	Thermo Fisher Scientific, Cat. No. 8094 or similar
Multimode or Luminescence Reader	discoverx.com/instrument-compatibility
Single and Multichannel Micro-pipettors and Pipette Tips	
50 mL and 15 mL Polypropylene Tubes	
1.5 mL Microtubes	
Tissue Culture Disposable Pipettes (1 mL-25 mL) and Tissue Culture Flasks (T25 and T75 flasks, etc.)	
Cryovials for Freezing Cells	
Hemocytometer	
Humidified Tissue Culture Incubator (37°C and 5% CO_2)	

* Series refer to the different sizes available for that reagent or kit.

Cell Vials and Handling Procedure

Two cryovials are shipped on dry ice and contain cells in 1 mL of AssayComplete™ Freezing Reagent (refer to the cell line-specific datasheet for the number of cells shipped in the vial). Upon receipt, the vials should be transferred to liquid nitrogen for storage beyond 24 hours. The following procedures are for the safe storage and removal of cryovials from liquid nitrogen storage.

1. Cells must arrive in a frozen state on dry ice.
2. Frozen cells must be transferred to either liquid nitrogen storage or a -80°C freezer immediately upon arrival. If cells will be thawed and used within 24 hours, they can be stored temporarily at -80°C. For longer storage, place vials in the vapor phase of liquid nitrogen storage.
3. When removing cryovials from liquid nitrogen storage, use tongs and place immediately on dry ice in a covered container. Wait at least one minute for any liquid nitrogen inside the vial to evaporate.
4. Proceed with the thawing and propagation protocols in the following section. Refer to the cell line product datasheet for the specific AssayComplete products listed in the protocol below.



Contact technical support immediately if cells are thawed upon arrival.



A face shield, gloves, and a lab coat should be worn during these procedures.



Cryovials should be stored in the vapor phase. They are not rated for storage in the liquid phase of liquid nitrogen.

Cell Culture Protocol

The following procedures are for thawing cells in cryovials, seeding and expanding the cells, maintaining the cell cultures once the cells have been propagated, and preparing them for assay detection.



Care should be taken in cell handling to avoid contamination.

Cell Thawing

The following is a protocol for thawing cells in a T75 flask.

1. Pre-warm AssayComplete Thawing Reagent in a 37°C water bath for 15 minutes.
2. Add 15 mL of the AssayComplete Thawing Reagent into a T75 flask in a sterile tissue culture hood. Set aside for **Step 4** below. DO NOT add selection antibiotics to the thawing reagent.



Safety Warning: A face shield, gloves, and a lab coat should be worn at all times when handling frozen vials. When removing cryovials from liquid nitrogen storage, use tongs and place immediately on dry ice in a covered container. Wait at least 1 minute for any liquid nitrogen inside the vial to evaporate. Do not touch the bottom of the vials at any time to avoid inadvertent thawing of the cells.

3. Remove the cell cryovials from -80°C or liquid nitrogen vapor storage and immediately place them on dry ice prior to thawing.
4. Place the frozen cell vials briefly (30 seconds to 1 minute) in a 37°C water bath, under sterile conditions, until only small ice crystals remain and the cell pellet is almost completely thawed. DO NOT centrifuge or vortex freshly thawed cells.
5. Decontaminate the vial by spraying and wiping with 70% ethanol, and transfer it to a tissue culture hood.

- With a pipette, gently transfer the thawed cells to the pre-filled T75 flask and incubate at 37°C and 5% CO₂.
- Maintain the cells in culture until they are >70% confluent. Then proceed to "Cell Propagation" instructions. Do not split if cells are below this confluency or growth issues may occur.

Cell Propagation

The following is a protocol for propagating cells once they become ≥70% confluent in a T75 flask.

- Pre-warm AssayComplete™ Thawing Reagent in a 37°C water bath for 15 minutes. DO NOT add selection antibiotics to the Thawing Reagent.
- Remove the T75 flask from the tissue culture incubator and place in a sterile tissue culture hood.
- Gently aspirate media from the T75 flask.
- Add 5 mL PBS into the T75 flask, very gently tip the flask side to side allowing PBS to cover the entire surface of the flask to rinse the cells.
- Gently aspirate PBS from flask.
- Add 1 mL of 0.25% Trypsin-EDTA to the T75 flask.
- Gently rock the flask back and forth to ensure the surface of the flask is completely covered with trypsin.
- Incubate the flask at 37°C, 5% CO₂ for 2 to 3 minutes or until the cells have detached.
- Remove the flask from the incubator and confirm the cells are detached by viewing under a microscope. If necessary, gently tap the edge of the flask to detach cells from the surface. If the cells do not detach, return to the incubator for an additional 1 to 2 minutes and repeat this step until cells are in suspension.
- Add 4 mL of AssayComplete Thawing Reagent to the T75 flask.
- Using a pipette, gently rinse the cells from the surface of the flask with the added media.
- Split the cells conservatively for the first passage after thaw using AssayComplete Thawing Reagent. Typical (conservative) split ratios for common cell backgrounds are included in the table below:



Prolonged treatment with Trypsin-EDTA may compromise cell viability

Cell Background	Suggested Split Ratio
CHO-K1	1:5
HEK 293	1:3
U2OS	1:2

For example, for CHO-K1 cells, add 4 mL of AssayComplete Thawing Reagent to the flask containing 1 mL of 0.25% Trypsin-EDTA, then transfer 1 mL (1/5th of the total reagent in the flask) into each new tissue culture flask.

- Fill a new T75 or T225 flask with AssayComplete Thawing Reagent until the total volume equals 12 mL for T75 flasks (or 45 mL for T225 flasks), and add cell suspension (added volume determined at [Step 12](#)) to the media in the flask. Transfer flask to a tissue culture incubator and incubate cells for 24 hours at 37°C and 5% CO₂.



To maintain logarithmic growth of the cells, cultures should be maintained in a sub-confluent monolayer.

14. After 24 hours, examine the cells under a microscope. If the cells appear to be healthy (adhering to the surface of the flask with only a small number of cells remaining in suspension) exchange the AssayComplete™ Thawing Reagent with AssayComplete Cell Culture Reagent supplemented with the recommended concentration of selection antibiotic (refer to cell line datasheet to determine the correct Cell Culture Kit and recommended antibiotic and antibiotic concentration for your cell line); 12 mL for T75 flasks (or 45 mL for T225 flasks). Then return the flask to a tissue culture incubator. If the cells do not appear to be healthy or if confluency is <25%, incubate for an additional 24 to 48 hours to allow for additional cell recovery before executing this step.
15. Once the cells become ≥70% confluent in the flask, split the cells every 2 to 3 days, based on the doubling time of the cell line, using AssayComplete Cell Culture Reagent supplemented with the proper selection antibiotics. Typical split ratios for common cell backgrounds are included in the table below:

Cell Background	Suggested Split Ratio
CHO-K1	1:10
HEK 293	1:5
U2OS	1:3

Cell Freezing

The following is a procedure for freezing cells that have been cultured in T75 or T225 flasks. This protocol assumes that cells have reached 70% to 80% confluency in enough flasks to fill the desired number of cryovials, in 1 mL/vial aliquots, with the desired number of cells per vial (e.g. 1 X 10⁶ per vial).

1. Remove T75 (or T225) flasks from incubator and place in a sterile tissue culture hood.
2. Gently aspirate the media from the flasks.
3. Add 10 mL PBS into each T75 flask (or 15 mL for a T225 flask), and swirl to rinse the cells.
4. Gently aspirate PBS from the flask.
5. Add 1 mL of 0.25% Trypsin-EDTA to T75 flasks (or 3 mL to T225 flasks).
6. Gently rock the flask back and forth to ensure the surface of the flask is completely covered with Trypsin-EDTA.
7. Incubate the flasks at 37°C and 5% CO₂ for 2 to 3 minutes or until the cells have detached.
8. Remove the flask from the incubator and view under a microscope to confirm that the cells have detached. If necessary, tap the edge of the flask to detach cells from the surface.
9. Add 5 mL AssayComplete Cell Culture Kit media to each T75 flask (or 15 mL to each T225 flask).
10. Using a pipette, gently rinse the cells from the surface of the flask with the added media. Gently pipette up and down several times to achieve a single cell suspension with no cell clumps.
11. Remove the entire amount of cells from the T75 flask and transfer to a 15 mL conical centrifuge tube. If using a T225 flask, transfer cells to a 50 mL conical centrifuge tube. If necessary, add additional AssayComplete Cell Culture Kit media to the flasks (e.g. 3 mL for T75 flasks or 5 mL for T225 flasks), rinse to collect the remaining cells, then transfer the additional media to the conical tube. Gently pipette up and down several times to ensure a single cell suspension.



Care should be taken in handling to avoid contamination.



Keep cells on ice during this process and transfer to a cryogenic container.

12. For the purpose of determining the concentration of cells in the suspension:
 - a. Set aside a small fraction (0.5 mL or less) of the suspended cells in a separate tube.
 - b. Transfer an appropriate portion of this fraction to a hemocytometer (typically 10 μ L of cell suspension) or another cell counting device.
 - c. Count cells, calculate the concentration of cells in the suspension, and then calculate the total number of cells remaining in the 15 mL (or 50 mL) centrifuge tube.
13. Centrifuge the collected cells at 300 X g for 4 minutes.
14. After centrifugation, discard the supernatant being careful not to disturb the cell pellet.
15. Based on the total cell number calculated in [Step 12](#) above, re-suspend cells to the desired concentration (e.g. 1×10^6 to 2×10^6 cells/mL) with ice cold AssayComplete™ Freezing Reagent.
16. Make 1 mL aliquots by transferring 1 mL of cell suspension to labeled 2 mL cryogenic tubes; tightly cap the tubes.
17. Freeze cells in a -80°C freezer at a controlled rate ($-1^{\circ}\text{C}/\text{minute}$) overnight. This can be performed using a dedicated cell freezer or commercially available freezing chamber (pre-chilled to 4°C). For short term storage, vials can be stored in the -80°C freezer for a maximum of two weeks.



Keep cells on ice during this process to protect cell viability.

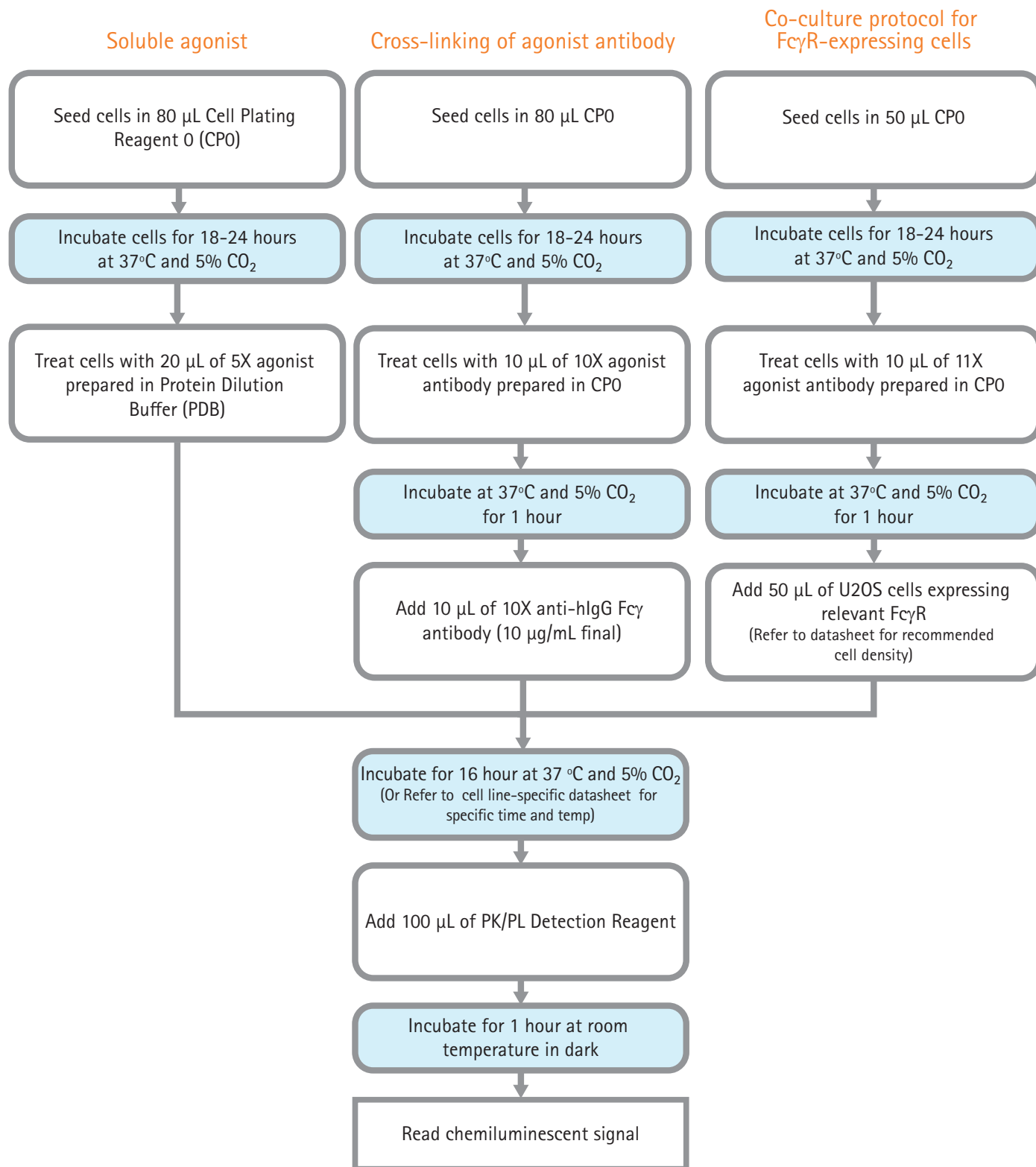
Protocol Schematic

Tip: Use this sheet to note your cell line-specific conditions. Post it on your bench, to use as a quick reference guide.

Assay Name: _____ Date: _____

Product Details: _____

Quick-Start Procedure: In a 96-well plate, perform the following steps.



Detailed Assay Protocol for Soluble Ligand or Antibodies

The following detailed protocol is specific for evaluating the ability of ligand or agonistic antibodies added as soluble reagents to activate co-stimulatory checkpoint receptors from the TNFR superfamily in a 96-well format cell-based assay

Note: For product-specific information on AssayComplete™ Cell Culture Media, Cell Plating Reagent, control ligand, incubation times and temperature, please see product datasheet.

Step 1: TNFRSF Signaling Cell Preparation and Plating

The following is a protocol for harvesting cells from a confluent T75 or T225 flask, and preparing them for plating in an assay plate at the desired concentration (optimal cell density). The protocol assumes that cells have been cultured in AssayComplete Cell Culture Media.



Avoid using trypsin for this cell assay preparation step.

Note: Do not use trypsin for this step; especially with assays interrogating membrane-anchored receptors (e.g. GPCRs), use of trypsin can negatively affect assay results.

1. Aspirate the media from the flask.
2. Add 2 mL AssayComplete Cell Detachment Reagent into each T75 flask (or 3 mL to T225 flasks), and swirl to rinse the cells.
3. Gently aspirate AssayComplete Cell Detachment Reagent from flask.
4. Add 1 mL fresh AssayComplete Cell Detachment Reagent to the T75 flasks (or 3 mL to the T225 flasks).
5. Gently rock the flask back and forth to ensure the surface of the flask is thoroughly covered with AssayComplete Cell Detachment Reagent.
6. Incubate at 37°C and 5% CO₂ for 2 to 5 minutes or until the cells have detached.
7. Remove the flask from the incubator and view under a microscope to confirm that the cells have detached. If necessary, tap the edge of the flask to detach cells from the surface.
8. Add 4 mL AssayComplete Cell Plating Reagent to T75 flasks (or 7 mL to T225 flasks). Note: Refer to datasheet to determine the correct Cell Plating Reagent for this cell line.
9. Using a pipette, gently rinse the cells from the surface of the flask with the added media.
10. Remove the entire amount of cells from the flask and transfer to a 15 mL conical centrifuge tube.
11. For the purpose of determining the concentration of cells in the suspension:
 - a. Set aside a small fraction (0.5 mL or less) of the suspended cells in a separate tube.
 - b. Transfer an appropriate portion of this fraction to a hemocytometer (typically 10 µL of cell suspension) or other cell counting device.
 - c. Count cells, then calculate the concentration of cells in the suspension, and then calculate the total number of cells remaining in the 15 mL centrifuge tube.
12. Centrifuge the collected cells at 300 x g for 4 minutes.

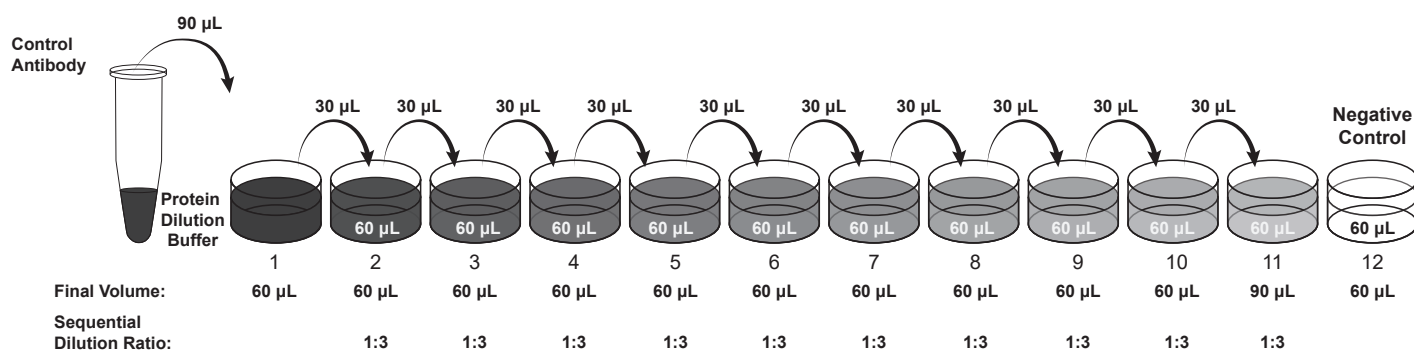
13. After centrifugation, discard the supernatant, and re-suspend the cell pellet in AssayComplete™ Cell Plating Reagent. Based on the cell number obtained in [Step 11 above](#), dilute the resuspended cells to the desired concentration (i.e. 250,000 cells/mL (5,000 cells/20 μ L).
14. Transfer 80 μ L/well of the cell suspension to a 96-well (or 20 μ L/well to a 384-well) assay plate.
15. Incubate the assay plate at 37°C and 5% CO₂. Refer to the cell line datasheet for recommended cell incubation time.
16. Proceed to ligand or antibody preparation and addition.

Step 2: Soluble Ligand or Antibody Preparation

When optimizing the assay conditions, it's recommended to include a positive control standard curve of the control ligand to verify that the kit components are working properly.

Prepare serial dilutions of the Control Ligand (standard) in a 96-well dilution plate by making a 3-fold, 11-point serial dilution in AssayComplete Protein Dilution Buffer (PDB) or appropriate ligand dilution buffer (as specified on the ligand datasheet). The 96-well dilution plate should be prepared at 5X of the final screening concentration.

1. For 11-point dilutions of control or test ligands, use a separate dilution plate (or non-binding tubes) and label the wells No. 1 to No. 12.
2. With a P200 pipettor, add 60 μ L of PDB to dilution wells No. 2 to No. 12. This is enough volume for 2 rows, and the dilution volume may be adjusted according to the number of wells desired.
3. Prepare the highest concentration of ligand in PDB (or ligand specific buffer). We recommend preparing the top dose of the ligand at a final screening concentration that is 250X the expected EC₅₀ (consult the dose response curve on the datasheet for the recommended top concentration for the control ligand). Therefore, prepare a working concentration that is 1250X the expected EC₅₀ per well to get a 5X working agonist concentration. The expected EC₅₀ concentration should lie near the center of the resulting dose response curve.
4. Add 90 μ L of the highest concentration of control ligand in PDB to well No. 1.
5. Remove 30 μ L of ligand from well No. 1 and add it to well No. 2. Mix gently.
6. With a clean tip, remove 30 μ L from well No. 2 and add it to well No. 3. Mix gently.
7. Continue this process through well No. 11. Do not add Control Ligand or Antibody to well No. 12 since this is the negative control well (containing only PDB).
8. Set up additional serial dilutions for additional ligands or soluble agonist antibodies in a similar manner using the ligand diluent recommended for the cell line on the cell line datasheet.



Control Antibody Serial Dilutions: Make 11 three-fold serial dilutions of control antibody in a separate dilution plate.

Step 3: Ligand Addition_____

- 1. Add 20 µL of each 5X ligand serial dilution from Step 2 to duplicate wells on top of the plated cells. (e.g. control ligand in rows A and B, test ligand in rows C and D as shown in the Representative Assay Plate Map).
- 2. Place lid back on plate and incubate assay plate at the specific temperature and for the time indicated on the cell line datasheet.

Step 4: Detection Reagent Addition_____

- 1. Prepare PathHunter PK/PL Detection reagent as described in the PathHunter PK/PL Detection reagent product insert (DiscoverX Cat. No. 93-0812).
- 2. Add 100 µL of PathHunter PK/PL Detection reagent to each well of the assay plate. Optional: Place the plate onto an orbital shaker at 350 rpm for 1 minute to cause even mixing.
- 3. Incubate assay plate for 1 hour at room temperature in the dark.



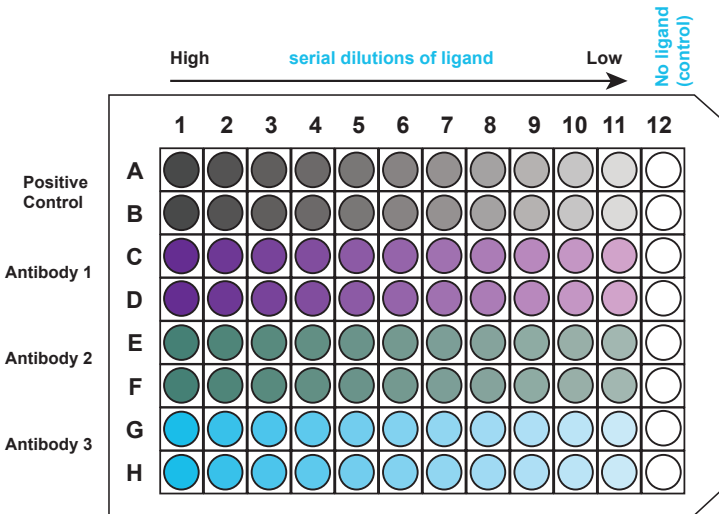
PathHunter PK/PL Detection Working Solution is light sensitive, thus incubation in the dark is necessary

Step 5: Assay Plate Reading_____

Read samples on a standard luminescence plate reader at 0.1 to 1 second/well for photomultiplier tube readers or 5 to 10 seconds for imager.

Data analysis can be performed using your choice of statistical analysis software (e.g. GraphPad Prism, Softmax Pro, Gen5, Microsoft Excel, etc.).

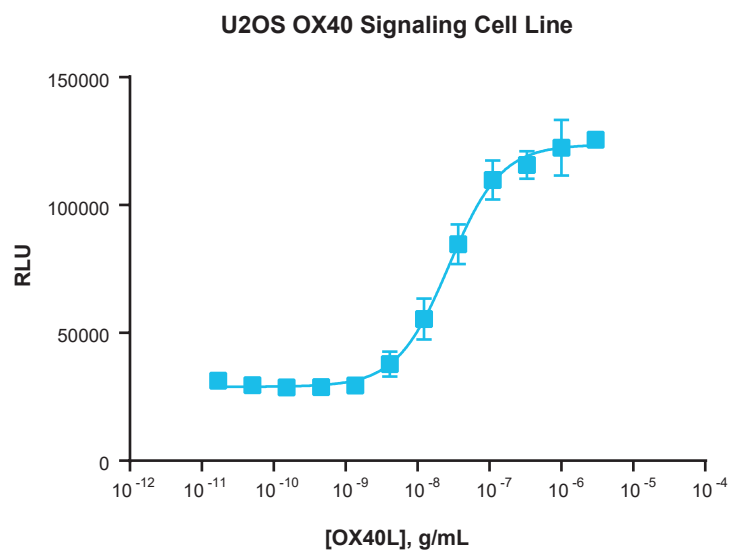
Representative Assay Plate Map



Assay Plate Map: This plate map shows an 11-point dose curve, with 3 data points at each concentration for one reference and one test sample per plate, with a variable serial dilution scheme.

Typical Results

Typical results are shown below for a PathHunter TNFRSF Checkpoint Assay using the protocol outlined for soluble ligands or antibodies. The data shows a potent, dose-dependent stabilization of the NIK protein using the OX40 Signaling Assay Cell Line in response to OX40 ligand.



EC ₅₀	27.4 ng/mL
S/B	4.0

Detailed Assay Protocol for Cross-linking Agonist Antibodies

The following detailed protocol incorporates cross-linking of agonistic antibodies with an anti-Fc antibody to enhance the activation of TNFRSF co-stimulatory checkpoint receptors in a 96-well format cell-based assay.

Note: For product-specific information on AssayComplete™ Cell Culture Media, Cell Plating Reagent, control ligand, incubation times and temperature, please see product data sheet.

Step 1: TNFRSF Signaling Cell Preparation and Plating

Prepare and plate assay cells as described in [Step 1](#) in the [Detailed Assay Protocol for Soluble Ligand or Antibodies](#) in 80 µL final volume per well.

Step 2: Agonist Antibody Preparation

Prepare serial dilutions of the agonist antibodies to be cross-linked as described for soluble antibodies in [Step 2](#) in the [Detailed Assay Protocol for Soluble Ligand or Antibodies](#) in a 96-well dilution plate by making a 3-fold, 11-point serial dilutions in AssayComplete Protein Dilution Buffer or appropriate ligand dilution buffer (as specified on the datasheet). The 96-well dilution plate should be prepared at 10X of the final screening concentration.

Step 3: Agonist Antibody Addition

1. Add 10 µL of each 10X agonist antibody serial dilution in duplicate to the plated cells according to the 96-well plate as indicated on the previously described assay plate map (e.g. Control Ligand in rows A and B; test antibodies in rows D and E).
2. Place lid back on plate and incubate assay plate for 1 hour at 37°C and 5% CO₂ incubator.

Step 4: Cross-linking Antibody Preparation

1. Prepare a fixed concentration (10 µg/mL final) of a human anti- IgG Fcγ antibody (e.g. RaPAb, Cat # 309-005-008, Jackson Immuno Research Labs) in Protein Dilution Buffer or AssayComplete Cell Plating O Reagent (CPO) as a 10X stock.
2. Add 25 µL per well to a new row of the master dilution plate for easy dispensing into the assay plate.

Step 5: Cross-linking Antibody Addition

1. Add 10 µL of the 10X stock of prepared cross-linking antibody to the appropriate wells of plated cells containing agonist antibody as indicated on the previously described assay plate map.
2. Place lid back on plate and incubate assay plate for remainder of assay incubation time at 37°C and 5% CO₂ incubator (refer to cell line datasheet for recommended incubation time). Optimal results for agonist antibodies are typically longer than required for soluble ligand (e.g. 16 hours for agonist antibodies).

Add detection reagents and read assay plates, as described in [Steps 4 and 5](#) in the [Detailed Assay Protocol for Soluble Ligand or Antibodies](#).

Detailed Assay Protocol for Cross-linking Agonist Antibodies by Co-culture with FcγR-expressing Cells

The following detailed protocol incorporates cross-linking of agonist antibodies in a physiologically relevant manner by co-culture with FcγR-expressing cells to enhance the activation of co-stimulatory checkpoint receptors in a 96-well format cell-based assay.

Step 1: TNFRSF Signaling Cell Preparation and Plating

Prepare and plate assay cells as described in [Step 1](#) in the [Detailed Assay Protocol for Soluble Ligand or Antibodies](#) in 50 μL final volume per well, at the density recommended in the cell line datasheet.

Step 2: Agonist Antibody Preparation

Prepare agonist antibody dilutions as in [Steps 2 and 3](#) in the [Detailed Assay Protocol for Cross-linking Agonist Antibodies](#), except that antibody dilutions should be prepared at 11X the desired final dose.

Step 3: FcγR-expressing Cells Preparation

Prepare a fixed concentration (e.g. 5K/50μl) of appropriate FcγR-expressing cells in AssayComplete Cell Plating 0 Reagent (CP0) as a 2X stock to achieve the recommended final density as per the cell line datasheet (e.g. typical range is 1.25K–2.5K/well of FcγRIIb expressing cells).

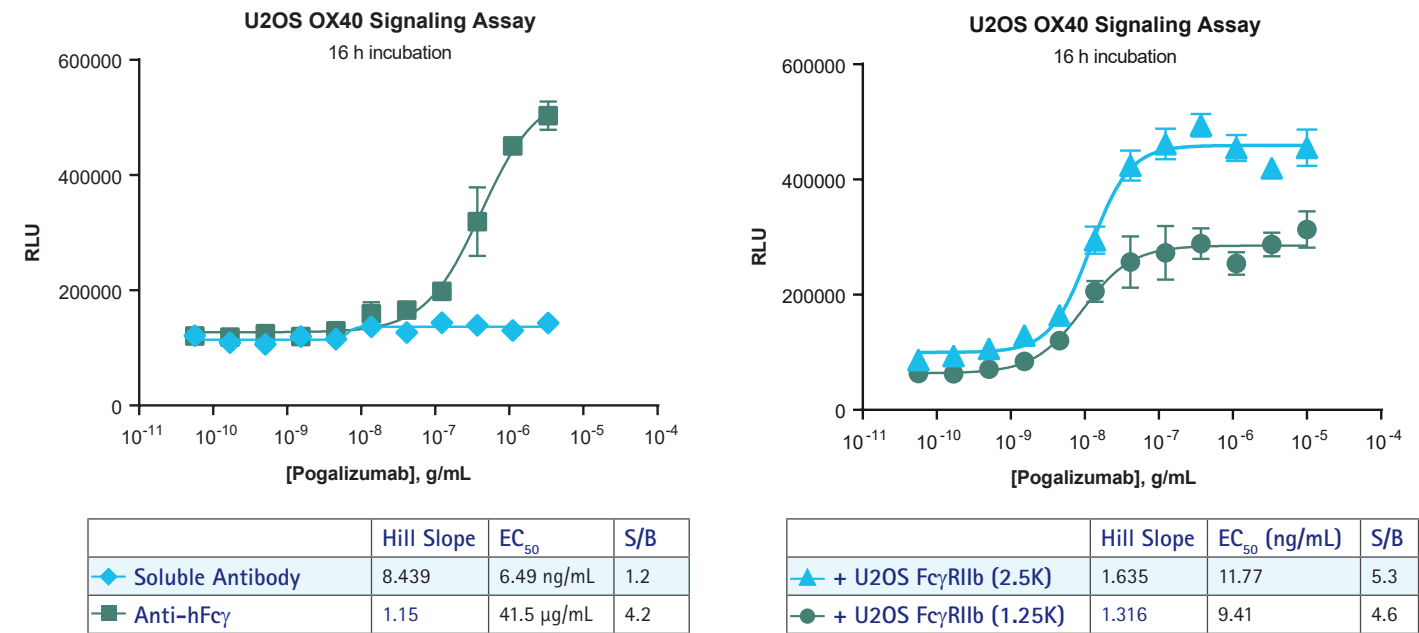
Step 4: FcγR-expressing Cells Addition

1. Add 50 μL of the 2X stock of prepared FcγR-expressing cells to the appropriate wells of plated cells containing agonist antibody as indicated on the previously described assay plate map.
2. Place lid back on plate and incubate assay plate for remainder of assay incubation time at 37°C and 5% CO₂ incubator (refer to cell line datasheet for recommended incubation time). Optimal results for agonist antibodies are typically longer than required for soluble ligand (e.g. 16 hours is optimal for most agonist antibodies).

Add detection reagents and read assay plates, as described in [Steps 4 and 5](#) in the [Detailed Assay Protocol for Soluble Ligand or Antibodies](#).

Typical Results

Typical results are shown below for a PathHunter TNFRSF Signaling Assay using the protocol outlined above for cross-linking antibodies. The data shows a potent, dose-dependent stabilization of the NIK protein using the OX40 Signaling Assay Cell Line in response to the OX40 antibody pogalizumab when cross-linked with an anti-human-Fcγ antibody, but not when antibody is added as a soluble reagent (A). Similarly, pogalizumab potently stimulates NIK accumulation in the OX40 Signaling Assay when co-cultured with FcγRIIb-expressing cells (B).



Troubleshooting Guide

Problem	Cause	Solution
No Response	Improper cell growth conditions	See datasheet for cell culture conditions.
	High DMSO/solvent concentration	Maintain DMSO/solvent at <1% in serial dilutions of compounds.
	Improper agonist used or improper agonist incubation time	See datasheet for recommended agonist and assay conditions.
	Improper preparation of agonist	Refer to specific datasheet to ensure proper handling, dilution, and storage of agonist.
	Improper time course for induction	Optimize time course of induction with control agonist.
Decreased Response	Higher passages give reduced performance	PathHunter cells are stable for at least 10 passages. Use low passage cells whenever possible.
Low or No Signal	Improper preparation of detection reagents	Detection reagents should be prepared just prior to use and are sensitive to light.
	Problem with cell growth, cell viability, or cell density	See datasheet for cell culture conditions.
	Problem with microplate reader	Microplate reader should be in luminescence mode. Read at 1 second/well.
Experimental S:B does not match datasheet value	Incorrect incubation temperature	Check and repeat assay at correct incubation temperature as indicated on the assay datasheet.
	Improper incubation time	Consult datasheet, and optimize incubation time with control agonist.
Cells growing slowly	U2OS grows slower than CHO-K1 or HEK 293	Doubling time is cell type specific, so please observe cells under microscope and monitor cell health.
	Slow growing clones	Use functionally-validated and optimized AssayComplete™ media and reagents to improve assay performance.
EC ₅₀ is right-shifted	1. Ligand not stored properly 2. Incorrect top dose for the dose response curve 3. Quality of ligand used in the assay	1. Follow directions for ligand storage as indicated on ligand datasheet. Store in small aliquots at -20°C. Do not freeze/thaw more than 1-2X. 2. Consult dose response curve on cell line datasheet for appropriate top concentration of ligand. 3. Differences in ligand vendors can also affect assay performance – be sure to use DiscoverX supplied ligand as a positive control to ensure that the assay works well in your hands before testing ligands or molecules from other sources.
	Problems with plate type and compound stability	Hydrophobic compounds should be tested for solubility and may be diluted in buffer containing 0.1% BSA. Non-binding surface plates may be necessary for hydrophobic compounds.
	Use of trypsin for cell plating	Use only AssayComplete Cell Detachment reagent to harvest cells for the assay.
High well-to-well variability in Z' study	Problems with plate type and compound solubility	Z' studies should be performed with automation.
		It may be necessary to test plate types and compound stability.

Limited Use License Agreement

These products may be covered by issued US and/or foreign patents, patent application and subject to Limited Use Label License.

Please visit discoverx.com/license for limited use label license and trademark information for this product.